

Cross-Type Semantic Alignment for Multi-Type Anomaly Detection in Heterogeneous Graphs

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Abstract

Graph Anomaly Detection (GAD) is critical in applications such as fraud prevention, cybersecurity, and social governance. While Graph Neural Networks (GNNs) have achieved remarkable success in detecting anomalies on homogeneous graphs, they face fundamental challenges in real-world heterogeneous settings involving diverse node types and imbalanced semantic richness. In heterogeneous graphs, nodes often vary significantly in semantic richness, with anomalies potentially spanning multiple types and emerging implicitly through cross-type dependencies. We identify two core limitations of existing methods: (i) the ineffective propagation of discriminative anomaly cues from informative to sparse nodes due to semantic imbalance, and (ii) conflicting optimization objectives arising from joint detection across multiple node types. To address these issues, we propose **CSA-MTHGAD**, a novel framework that integrates smoothness-guided cross-type semantic alignment with dynamic multi-task learning. It selectively propagates anomaly-sensitive features across types and harmonizes task-specific gradients through adaptive projection and weighting. To facilitate research, we employ two real-world heterogeneous benchmarks in the domain of social governance. Extensive experiments demonstrate that CSA-MTHGAD achieves superior performance over state-of-the-art baselines in accuracy, robustness, and generalization for multi-type anomaly detection.

CCS Concepts

• **Security and privacy** → **Intrusion/anomaly detection and malware mitigation**; *Web application security*.

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Keywords

Anomaly Detection; Heterogeneous Graph; Social Governance

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1 Introduction

Graphs offer a powerful representation paradigm for relational data by explicitly modeling entities and their interactions as nodes and edges. Such representations have been widely adopted across a range of graph learning tasks, including fake news detection [13, 16], social behavior analysis [9, 35], and sentiment analysis [17, 30, 31]. These diverse applications demonstrate the capacity of graph models to capture complex dependencies beyond independent and identically distributed assumptions. Within this general setting, anomaly detection has emerged as a particularly important problem, as identifying rare and irregular patterns is crucial for maintaining system reliability and security. In practice, graph anomaly detection has been successfully applied to domains such as financial fraud prevention [1, 5, 38] and network intrusion monitoring [12, 23]. Recent advances in graph-based learning have further promoted this research direction. By learning expressive representations that integrate both structural context and node attributes, Graph Neural Networks (GNNs) [11, 26, 34] have significantly improved anomaly detection performance on graph data. In homogeneous graphs, GNN-based methods, especially those leveraging reconstruction or contrastive learning objectives [15, 22, 40], have achieved impressive performance.

However, real-world systems, especially those arising in social governance scenarios, are inherently heterogeneous, involving multiple types of entities and relations. As illustrated in Figure 1(a), a social governance network typically consists of person, event, and

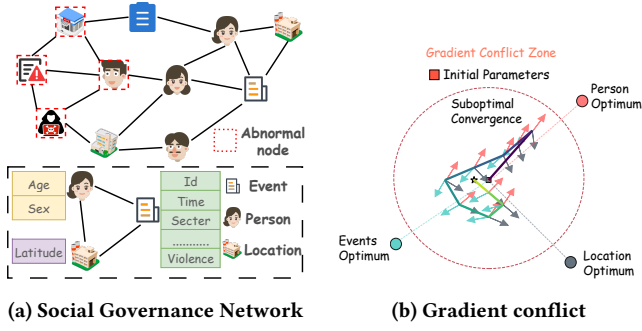


Figure 1: (a) is a social governance network. (b) reveals schematic diagram of gradient conflicts in the parameter space of multi-type node anomaly detection.

location nodes, where anomalies may correspond to individuals repeatedly involved in violent behaviors, abnormal events such as cybercrime or fraud, or suspicious locations with unusually high incident density. These node types exhibit markedly different feature characteristics: event nodes are often text-rich and provide explicit semantic descriptions of risks, whereas person nodes are text-less and usually contain only sparse demographic information. Notably, text-less nodes may appear normal when considered in isolation, yet implicitly convey abnormality through their connections to high-risk entities. This stands in contrast to homogeneous graphs, where anomalies are commonly assumed to manifest as explicit deviations. In heterogeneous graphs, anomalies are often subtle, indirect, and distributed across node types, rendering detection considerably more challenging.

Despite the practical significance, anomaly detection in heterogeneous graphs remains largely underexplored. Most existing methods target homogeneous settings [3, 9, 21, 33], where abnormal nodes are assumed to exhibit distinguishable features [7, 20]. The few attempts to incorporate heterogeneity often rely on naive extensions, such as replacing homogeneous encoders with heterogeneous GNN variants [37], without addressing the intrinsic cross-type semantic inconsistencies and imbalance in anomaly signal richness. As a result, informative cues from text-rich nodes may be diluted during propagation to text-less nodes, weakening their effectiveness. This creates the first key challenge: *how to extract, preserve, and propagate highly discriminative anomaly signals across node types with varying semantic density while avoiding signal dilution or masking.*

A second fundamental challenge arises from the multi-type nature of anomalies in heterogeneous graphs. An effective detection framework must jointly capture abnormal patterns across all node categories, rather than focusing on a single type. However, existing approaches often treat anomaly detection as a single-task problem, either optimizing for one node category or training each type in isolation. This neglects interdependence across types and leads to conflicting optimization objectives during joint learning. As illustrated in Figure 1(b), gradients from person, event, and location nodes may point in divergent directions, pushing model parameters toward incompatible optima and resulting in suboptimal convergence. Such gradient conflicts hinder generalization and highlight

the second challenge: *how to design a coordinated multi-task optimization strategy that harmonizes conflicting learning signals and enables stable, collaborative anomaly detection across types.*

To tackle these challenges, we propose **CSA-MTHGAD** (Cross-Type Semantic Alignment for Multi-Type Heterogeneous Graph Anomaly Detection), a unified framework tailored to the complex demands of heterogeneous graphs. CSA-MTHGAD integrates semantic alignment and anomaly signal enhancement with multi-task optimization, enabling collaborative and type-aware detection in real-world heterogeneous graphs. Specifically, the semantic alignment module conducts fine-grained smoothness analysis [19] on text-rich nodes to identify anomaly-sensitive, low-smoothness features, which are selectively propagated via heterogeneous message passing to text-less nodes. This enables sparse nodes to inherit meaningful cues from high-information neighbors. In parallel, the multi-task anomaly detection module monitors per-task learning dynamics and applies gradient projection and adaptive weighting to mitigate inter-task conflicts, ensuring stable optimization and improving generalization across node types. Our key contributions are summarized as follows:

- **Problem Setting.** We present and analyze a systematic study of multi-type anomaly detection in complex heterogeneous graphs, clearly highlighting the crucial challenges of semantic imbalance and cross-type dependencies that have been largely overlooked in prior work.
- **Methodology.** We introduce CSA-MTHGAD, a novel framework that combines cross-type semantic alignment and anomaly signal enhancement with a task-aware gradient harmonization strategy, enabling accurate and collaborative anomaly detection across diverse node types.
- **Empirical Validation.** Experiments on real-world social governance graphs demonstrate that CSA-MTHGAD consistently and significantly outperforms state-of-the-art baselines by a considerable margin, exhibiting superior accuracy, robustness, and generalization capabilities in multi-type anomaly detection.

2 Preliminary

2.1 Problem Statement

The goal of this study is to address graph anomaly detection in heterogeneous graphs, where nodes and edges belong to multiple types. A heterogeneous graph is denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{R}, \phi, \psi)$, where $\mathcal{V}$ and $\mathcal{E}$ are the sets of nodes and edges, $\mathcal{T}$ and $\mathcal{R}$ represent node and edge types, and $\phi : \mathcal{V} \rightarrow \mathcal{T}, \psi : \mathcal{E} \rightarrow \mathcal{R}$ are type-mapping functions. In the context of social governance, we distinguish text-rich nodes ($\mathcal{T}_{rich}$) from text-less nodes ($\mathcal{T}_{less}$). A meta-path [10, 39, 42] is defined as a semantic relation sequence between node pairs; for example, the PEP (Person–Event–Person) path captures the connection between two individuals through shared events. For a given meta-path m , we define a view $g^{(m)} = \{\mathcal{X}, A^{(m)}\}$, where $A^{(m)}$ is the adjacency matrix under path m and $\mathcal{X}$ is the shared attribute space.

In our setting, nodes linked to elevated risk cases are regarded as anomalous, while the majority remain normal. This naturally produces the typical imbalance $|\mathcal{V}_a| \gg |\mathcal{V}_n|$ in anomaly detection tasks. Each node $v \in \mathcal{V}$ is thus assigned a binary indicator

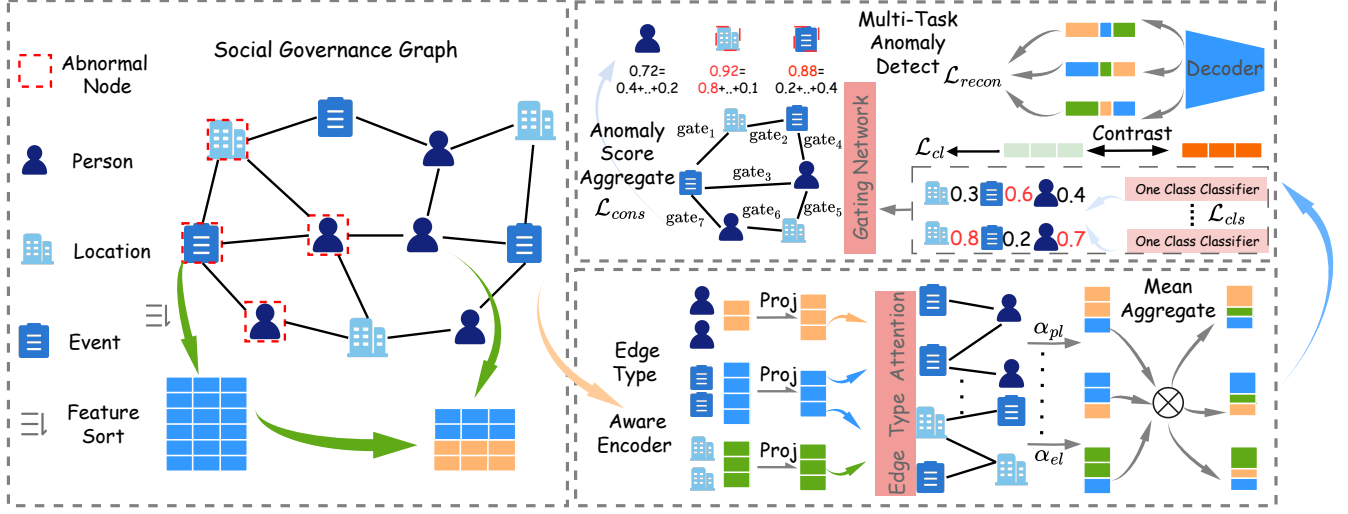


Figure 2: CSA-MTHGAD overview: The input social-governance graph contains text-rich EVENT nodes (blue) that emit explicit anomaly cues (red). Their low-smoothness features are propagated to neighboring text-less PERSON and LOCATION nodes, creating enriched encoder inputs. The edge-type-aware encoder first applies attention to preserve relation semantics, then mean-aggregates to fuse local evidence and projects nodes into embeddings. A multi-task anomaly head integrates per-type classifiers, a gating network for score sharing, attribute reconstruction, and contrastive learning; gradient coordination aligns these tasks and exposes hidden anomalies in every node type.

$y \in \{0, 1\}$, with 1 representing anomalous (high-risk) nodes and 0 representing normal ones. The objective is to perform supervised anomaly detection for heterogeneous node types: for each type $\tau \in \mathcal{T}$, we learn a function $f_\tau : \mathcal{V}_\tau \rightarrow [0, 1]$ that estimates the anomaly probability for nodes of type τ .

3 Methodology

In this section, we present a comprehensive description of the proposed CSA-MTHGAD model. The overall framework of CSA-MTHGAD is illustrated in Figure 2. We first introduce the overall architecture, emphasizing how its constituent modules collaboratively address the core challenges inherent in social governance networks. Subsequently, we highlight two core innovations that underpin the effectiveness of the proposed method.

We begin by introducing the Cross-Type Feature Alignment module, which is designed to bridge the information gap between diverse node types. Specifically, anomaly-sensitive features from text-rich nodes are selected based on low feature smoothness and selectively propagated to text-less nodes. Optimize synthesized augmented features via lightweight fusion and ensure consistent propagation dynamics through regularization. We then describe the Multi-Task Anomaly Detection module, which facilitates stable and efficient joint optimization for anomaly detection across heterogeneous node types. This module integrates semantic embeddings into a unified representation space using a two-layer heterogeneous graph neural network, capable of capturing structural and semantic dependencies. Anomaly detection is enhanced through attribute reconstruction, contrastive learning, and gated anomaly score propagation. A task-aware gradient harmonization strategy

further ensures balanced convergence across multiple anomaly detection tasks during the training process.

3.1 Cross-Type Feature Alignment

3.1.1 Low Smoothness Feature Selection. Prior research has demonstrated that low feature smoothness tends to indicate drastic variations across local neighborhoods, which are often strongly correlated with anomalous[19]. Motivated by this observation, we calculate the smoothness of each feature dimension across adjacent nodes to select the most anomaly-relevant features for the $\mathcal{T}_{rich}$ nodes as follows:

$$S_{rich}(k) = -\frac{1}{|\mathcal{E}_{rich}|} \sum_{(u,v) \in \mathcal{E}_{rich}} (x_{u,k} - x_{v,k})^2, \quad (1)$$

where $\mathcal{E}_{rich}$ contains edges incident to at least one rich node. We retain the K smallest entries and realise the selection through $X_{sel}^{rich} = X^{rich}P$, with P a binary column filter. These anomaly sensitive dimensions will be propagated to text-less nodes next.

3.1.2 Selective Feature Propagation. In the Selective Feature Propagation stage, edge attention guides the injection of selected features from text-rich nodes into their connected text-less nodes. For an edge (i, j, r) with relation type r , we compute type-specific features h_i for node i and h_j for node j as follows:

$$h_i = \text{ReLU}(\mathbf{W}_{\phi(i)} \mathbf{x}_i), \quad h_j = \text{ReLU}(\mathbf{W}_{\phi(j)} \mathbf{x}_j), \quad (2)$$

where $\mathbf{W}_{\phi(i)} \mathbf{x}_i$ and $\mathbf{W}_{\phi(j)} \mathbf{x}_j$ are learnable transformation matrices for the respective node types. Edge attention $\alpha_{ij}^{(r)}$ is then calculated as:

$$\alpha_{ij}^{(r)} = \frac{\exp(\mathbf{a}^\top \tanh([\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{e}_r]))}{\sum_{k \in \mathcal{N}_j^{(r)}} \exp(\mathbf{a}^\top \tanh([\mathbf{h}_k \parallel \mathbf{h}_j \parallel \mathbf{e}_r]))}, \quad (3)$$

where $\mathbf{e}_r$ is the edge type embedding, and $\mathcal{N}_j^{(r)}$ denotes the set of r -type neighbors of node j . Finally, all incoming messages are aggregated to form the final message $\mathbf{m}_j$ at the target node j :

$$\mathbf{m}_j = \sum_{r \in \mathcal{R}} \sum_{i \in \mathcal{N}_j^{(r)}} \alpha_{ij}^{(r)} \mathbf{h}_i. \quad (4)$$

3.1.3 Fusion and Lightweight Enhancement. The aggregated message is first concatenated with the original node attribute to form an enhanced input representation, calculated as:

$$\tilde{\mathbf{x}}_j = f_{\text{MLP}}([\mathbf{x}_j \parallel \mathbf{m}_j]). \quad (5)$$

Each $\mathcal{T}_{\text{less}}$ node applies a multi-head self-attention block, followed by a feed forward network to refine the features:

$$\mathbf{Z}_t = \text{MHA}(\tilde{\mathbf{X}}_t) + \tilde{\mathbf{X}}_t, \quad \mathbf{X}'_t = \text{FFN}(\text{LN}(\mathbf{Z}_t)) + \mathbf{Z}_t. \quad (6)$$

Hence $\mathbf{X}'_t$ joins explicit signals from rich neighbours with the original sparse description, enabling implicit anomalies to surface.

3.1.4 Propagation Regularity. To limit distortion of dominant rich dimensions, we define $\mathcal{L}_{\text{prop}}$ that encourages type-specific propagation operators to maintain orthogonality:

$$\mathcal{L}_{\text{prop}} = \sum_{t \in \mathcal{T}_{\text{less}}} \|(\mathbf{M}_t \mathbf{W}_t)(\mathbf{M}_t \mathbf{W}_t)^\top - \mathbf{I}\|_F^2, \quad (7)$$

where $\mathbf{M}_t \mathbf{W}_t$ represents the feature propagation matrix from $\mathcal{T}_{\text{rich}}$ nodes to target-type nodes, parameterized by a learnable weight matrix $\mathbf{W}_t$. To preserve orthogonality in propagation relationships, we regularize $\mathbf{M}_t \mathbf{W}_t$ to approximate the identity matrix $\mathbf{I}$. Minimizing this regularization ensures efficient feature transmission while preventing the degradation of information during node representation propagation.

3.2 Multi-Task Anomaly Detection

Let $\tilde{\mathbf{x}}_v$ denote each node's aligned feature. We use a two-layer heterogeneous GNN that first preserves relation heterogeneity and then consolidates local evidence.

3.2.1 Selective Aggregation. For each node v , we compute relation-wise first-layer summaries $\mathbf{h}_{v,r}^{(1)}$ as:

$$\mathbf{h}_{v,r}^{(1)} = \text{ReLU}\left(\sum_{u \in \mathcal{N}_v^{(r)}} \beta_{uv}^{(r)} \mathbf{W}_r^{(1)} \tilde{\mathbf{x}}_u\right), \quad (8)$$

where $\beta_{uv}^{(r)}$ are relation-specific attention coefficients normalized over $u \in \mathcal{N}_v^{(r)}$ (including the self-loop $u=v$), and $\mathbf{W}_r^{(1)}$ is the relation-specific weight. And a consolidated second-layer representation $\mathbf{h}_v^{(2)}$ as:

$$\mathbf{h}_v^{(2)} = \text{ReLU}\left(\sum_{r \in \mathcal{R}} \gamma_v^{(r)} \mathbf{W}^{(2)} \mathbf{h}_{v,r}^{(1)}\right), \quad (9)$$

where $\gamma_v^{(r)} = 1/|\mathcal{R}|$ gives uniform across-relation fusion and $\mathbf{W}^{(2)}$ is the second-layer weight. Finally, with projection $\mathbf{P}$, we set $\mathbf{z}_v = \mathbf{P}\mathbf{h}_v^{(2)}$ for anomaly detection.

3.2.2 Attribute Reconstruction. To keep fine-grained semantics that help reveal implicit anomalies we decode node attributes and optimize the following objective:

$$\hat{\mathbf{x}}_v = \mathbf{D}_{\phi(v)} \mathbf{z}_v, \quad \mathcal{L}_{\text{rec}} = \sum_v \|\hat{\mathbf{x}}_v - \mathbf{x}_v\|_2^2. \quad (10)$$

3.2.3 Contrastive Learning. Neighbourhood-level learning further sharpens the embedding. For every observed edge (u, v) we define

$$\mathcal{L}_{uv}^{\text{cl}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_u, \mathbf{z}_v)/\tau)}{\sum_{v' \neq v} \exp(\text{sim}(\mathbf{z}_u, \mathbf{z}_{v'})/\tau)}, \quad (11)$$

and aggregate $\mathcal{L}_{\text{cl}} = \sum_{(u,v) \in \mathcal{E}} \mathcal{L}_{uv}^{\text{cl}}$. Here, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, $\tau > 0$ is a temperature. This promotes consistency with the local neighbourhood structure in the embedding.

For negative sampling, we randomly select unconnected node pairs (z_{-u}, z_{-v}) from different node types (e.g., between 'person' and 'events' nodes). These negative samples are used to push the model to distinguish between similar and dissimilar nodes, improving the quality of the embeddings by ensuring a clear separation between positive and negative samples.

3.2.4 Anomaly Scoring and Propagation. We compute node-wise logits $\mathbf{o}_v$ and the primary anomaly score s_v^{orig} as

$$\mathbf{o}_v = \mathbf{W}_{\phi(v)}^{\text{cls}} \mathbf{z}_v, \quad s_v^{\text{orig}} = \text{softmax}(\mathbf{o}_v)[1], \quad (12)$$

where $\mathbf{W}_{\phi(v)}^{\text{cls}}$ is the type-specific classifier and $[1]$ indexes the anomaly class. To propagate anomaly evidence along typed edges (u, v, r) , we define an edge-wise gate and the propagated score:

$$g_{uv}^{(r)} = \sigma(\mathbf{w}^\top [\mathbf{h}_u^{(2)} \parallel \mathbf{h}_v^{(2)}]), \quad (13)$$

$$s_v^{\text{prop}} = \frac{\sum_{(u,v,r)} g_{uv}^{(r)} s_u^{\text{orig}}}{\sum_{(u,v,r)} g_{uv}^{(r)} + \epsilon}.$$

Here $\sigma(\cdot)$ is the logistic function and $[\cdot \parallel \cdot]$ denotes concatenation. The propagation is a gate-weighted average over incoming neighbors, where $g_{uv}^{(r)}$ down-weights unreliable links and the denominator stabilizes the scale across nodes. To mitigate hub amplification, we optionally apply degree correction by setting $g_{uv}^{(r)} \leftarrow g_{uv}^{(r)} / \sqrt{d_u d_v}$, where d_u and d_v are node degrees. Finally, we enforce consistency between the primary and propagated scores via

$$\mathcal{L}_{\text{cons}} = \sum_v (s_v^{\text{orig}} - s_v^{\text{prop}})^2. \quad (14)$$

3.2.5 Task Aware Gradient Coordination. Let $\mathcal{L}_{\text{cls}}^t$ denote the cross entropy loss for node type t and n_t its training size. We compute dynamic weights as:

$$\pi_t = \frac{\mathcal{L}_{\text{cls}}^t}{\sum_{t'} \mathcal{L}_{\text{cls}}^{t'}} \frac{1}{\sqrt{n_t + \epsilon}}, \quad (15)$$

where n_t is the number of nodes of type t , and ϵ is a small constant to avoid zero errors. The final classification loss, adjusted for multi-task learning is $\tilde{\mathcal{L}}_{\text{cls}} = \sum_t \pi_t \mathcal{L}_{\text{cls}}^t$.

Besides, to prevent overfitting and ensure efficient task learning, we monitor the performance of each task’s classifier on the validation set. If the performance of a classifier for a particular task saturates (i.e., it reaches a stable value), we freeze the classifier’s parameters, so the network pays more attention to unsolved tasks. And if gradients of two tasks conflict, that is $\langle g_t, g_{t'} \rangle < 0$, we project

$$g_t \leftarrow g_t - \frac{\langle g_t, g_{t'} \rangle}{\|g_{t'}\|^2} g_{t'}, \quad (16)$$

by applying task-aware gradient coordination, we ensure that each node type’s gradients are adjusted to maintain balance during the backpropagation process, allowing for better convergence in multi-task optimisation.

3.2.6 Unified Objective. We have the complete objective as

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{rec}} + \lambda_2 \mathcal{L}_{\text{cl}} + \lambda_3 \tilde{\mathcal{L}}_{\text{cls}} + \lambda_4 \mathcal{L}_{\text{cons}} + \lambda_5 \mathcal{L}_{\text{prop}}. \quad (17)$$

Hyper-parameters λ_* are balanced parameters. Based on this loss, we amplify signals from text-rich nodes and reveal hidden anomalies on text-less nodes, while gradient coordination ensures balanced convergence across all node types.

Algorithm 1 CSA-MTHGAD: Cross-Type Semantic Alignment for Multi-Type Anomaly Detection

- 1: **Input:** Heterogeneous graph $G = (V, E, T, R, \phi, \psi)$, node features, edge types, hyperparameters K, L_{rec} , and L_{cl} .
- 2: **Cross-Type Feature Alignment:**
- 3: Select anomaly-sensitive features from text-rich nodes based on low smoothness, resulting in $\tilde{X}_{\text{rich}}$.
- 4: Propagate features to text-less nodes using edge attention, forming enhanced node representations X_{enhanced} .
- 5: **Multi-Task Anomaly Detection:**
- 6: Compute type-specific features for each node v_i using heterogeneous GNN, resulting in embeddings h_i .
- 7: Aggregate the relation-wise features across multiple node types $r \in R$ to form $h_{(2)}$.
- 8: Perform anomaly detection using attribute reconstruction L_{rec} and contrastive learning L_{cl} .
- 9: Compute anomaly scores s_{orig} for each node v_i based on the anomaly class probability.
- 10: **Gradient Coordination:**
- 11: Apply task-aware gradient coordination to harmonize multi-task learning signals and resolve gradient conflicts across node types.
- 12: **Test Stage:** Given test node v_i , compute the final anomaly score $s_{\text{final}}(v_i)$.

3.3 Computational Complexity

Let N_t and F_t denote, respectively, the number and input-feature dimension of nodes of type t , and let $E_{t \rightarrow s}$ be the number of directed edges of type $(t \rightarrow s)$; we write $E = \sum_{t \rightarrow s} E_{t \rightarrow s}$ and H for the unified hidden dimension. The one-off preprocessing step (*feature-smoothness evaluation*) visits each homogeneous event–event edge exactly once, yielding a cost of $O(E_{\text{homo}} F_{\text{events}})$. During training, a single forward–backward pass comprises: (i) multi-head self-attention-based node enhancement with cost $\sum_t O(N_t F_t H + N_t H^2)$;

(ii) cross-type feature propagation whose attention and scatter-add operations cost $\sum_{t \rightarrow s} O(E_{t \rightarrow s} H)$; and (iii) stacked GAT and SAGE convolutions which dominate the iteration with $O((h+1)EH)$ time and $\Theta(EH + NH)$ activation memory (h heads). Hence the per-iteration wall-clock complexity is $\Theta((h+1)EH)$, linear in the number of edges, while the total training cost over T epochs is $T\Theta((h+1)EH)$, up to an additive evaluation term. Because all edge operations are streamed and no dense adjacency is materialised, memory also scales linearly with E and N , establishing the algorithm’s suitability for million-node, multi-million-edge heterogeneous graphs.

4 Experiments

4.1 Experimental Setup

4.1.1 Dataset. We evaluate our approach on two heterogeneous social governance graphs derived from real-world risk cases. Each graph involves four categories of nodes: *person*, *location*, *event*, and *event-type*. The first three correspond to directly observed entities, while event-type nodes capture abstract categories of incidents (e.g., telecom fraud, cyber espionage), enabling the identification of broader patterns once specific events are linked to them. The first dataset primarily reflects traditional governance scenarios such as disputes and violent incidents, whereas the second dataset further incorporates emerging risks such as online fraud and cyber espionage, thereby offering a richer and more diverse benchmark.

All sensitive information is anonymized in order to protect personal privacy; for example, actual place names are replaced with unique identifiers. Risk levels available in the source records are used to define anomaly labels: nodes associated with high-risk cases are regarded as anomalous, while the rest are treated as normal. This leads to a highly imbalanced label distribution, consistent with realistic anomaly detection tasks where anomalies are rare compared with normal instances.

These datasets are particularly suitable for anomaly detection research for several reasons. First, they are inherently heterogeneous: entities of different types coexist and interact through multiple relation types, forming complex semantic structures. Second, the anomaly signals are unevenly distributed across node types. For example, events are text-rich and contain explicit descriptions of risky activities, while persons and locations are text-less and only provide limited demographic or geographic attributes. This imbalance makes it challenging to propagate useful anomaly cues from informative nodes to sparse ones. Third, anomalies may not always be evident within a single type of node; instead, they often emerge implicitly through cross-type dependencies, such as suspicious individuals being connected to abnormal events or high-risk locations. These properties closely mirror real-world scenarios, making the datasets well aligned with the objectives of multi-type anomaly detection.

The first dataset contains 2,490 nodes and 8,736 edges, while the second comprises 30,395 nodes and 119,140 edges. Detailed statistics of the two benchmarks, including the distribution of node types and anomalies, are summarized in Table 1.

Dataset	Social-Gover I	Social-Gover II
Person	1051	12024
Event	1092	15277
Location	301	2931
Event-Type	46	163
Anomalies	445	3132
Ratio	17.87%	10.49%

Table 1: Statistics of two social governance benchmarks.

4.1.2 Baselines. To our best knowledge, there exist no supervised GAD methods specifically designed for multi-type node anomaly detection on heterogeneous graphs. To assess CSA-MTHGAD rigorously, we construct two baseline groups: (i) We first adopt heterogeneous graph representation methods to encode complex node dependencies, followed by AutoEncoder-based anomaly detectors and an additional classification module for supervised training. Specifically, we select three representative methods: HAN[32], HGT[14], HGRAN[29] and HGATE[28]. To align these methods with our supervised anomaly detection setting, we incorporate an AutoEncoder into each model to reconstruct node features and structures, using the reconstruction errors as anomaly scores. Subsequently, a classifier is appended to leverage available labels. (ii) Due to the absence of dedicated multi-type heterogeneous GAD baselines, we establish a set of weaker baselines by transforming our heterogeneous social governance graph into a homogeneous counterpart. This conversion enables the application of established homogeneous GAD methods. Despite the homogeneity assumption during training, the node-type information remains accessible at evaluation time, allowing us to separately assess the anomaly detection performance on each node type. Similar to the previous category, these methods are augmented with a classifier and optimized via cross-entropy loss. The selected homogeneous GAD methods include: DOMINANT[7], CoLA[20], TAM[22], AEGIS[6] and TFGAD[41]. These homogeneous baselines serve as meaningful comparisons, highlighting the necessity and advantages of explicitly modeling graph heterogeneity and multi-type anomaly interactions as addressed by CSA-MTHGAD.

4.1.3 Evaluation and Implementation. We adopt AUC, Macro-F1, and Micro-F1[4, 25, 36] as our evaluation metrics. AUC is a standard metric widely used in prior graph anomaly detection studies. Macro-F1 evaluates the average performance across all node types, treating each type equally. Micro-F1 captures the overall detection quality by accounting for imbalanced node distributions in the heterogeneous graph. We implement our model and all baselines using PyTorch and DGL[27]. All methods are trained under the same settings: the number of training epochs is 300, with a dropout rate of 0.5 and a weight decay of $5e - 4$. We use the Adam optimizer with a learning rate of $1e - 3$ and a batch size of 256. The embedding dimension is fixed to 128 for all models, and early stopping is applied with a patience of 20 epochs. During training, we identify anomaly-sensitive dimensions on text-rich nodes via local feature smoothness analysis and inject the top- K low-smoothness

features into neighboring text-less nodes through heterogeneous message passing. Rather than fixing K , we parameterize it by a ratio `keep_ratio` defined as $K/|\mathcal{F}_{\text{rich}}|$, where $|\mathcal{F}_{\text{rich}}|$ denotes the number of feature dimensions on text-rich nodes. We set `keep_ratio` to 0.4. To mitigate gradient conflicts and overfitting during multi-type optimization, we dynamically freeze the classifier of each node type if its validation AUC shows no improvement for 5 consecutive evaluations. For homogeneous baselines we apply principal component analysis (PCA)[24] to unify feature dimensions across different node types, and discard edge-type information to create a simplified homogeneous graph.

4.2 Performance Comparison

The experimental results of baseline models and our proposed CSA-MTHGAD model in the multi-type anomaly detection task are demonstrated in the Table 2. We summarize several key observations:

CSA-MTHGAD attains the best (or tied best) AUC on all node types under both Social-Gover I and Social-Gover II. This consistent advantage indicates that our cross-type semantic alignment plus anomaly feature propagation pipeline produces more discriminative ranking scores between anomalous and normal nodes, which is the primary objective in anomaly detection.

In a few columns certain baselines yield slightly higher Macro-F1 or Micro-F1. There are three possible reasons: (i) These methods implicitly optimize decision boundaries for balanced multi-class classification, whereas CSA-MTHGAD emphasizes separating anomalous from normal nodes in the score space rather than aggressively tuning a fixed threshold for per-class F1. (ii) Macro-F1 is sensitive to per-class (here per risk level) calibration on minority normal sub-categories within a node type; some baselines that underfit the anomaly signal produce smoother probability distributions, occasionally yielding a marginally higher harmonic mean of precision and recall even though their ranking quality is poorer. (iii) Micro-F1, being dominated by frequent node types / majority labels, can favor models that bias toward the prevalent “non-anomalous” class; our model deliberately increases recall on true anomalies at the cost of a small precision drop on abundant normal instances, which can slightly reduce Micro-F1. Thus, higher F1 without higher AUC typically reflects threshold / calibration effects rather than better anomaly discrimination.

The AUC improvements occur for Person and Location nodes, which inherently possess sparse attribute profiles. These gains confirm that our approach of integrating distilled low-smoothness Event features—direct indicators of anomalies, while coordinating multi-task optimization synergistically elevates representation quality precisely where conventional methods underperform.

In summary, CSA-MTHGAD delivers the most reliable anomaly ranking ability across all node categories; slight F1 advantages of some baselines stem from calibration biases rather than genuinely superior discrimination. This validates our design focus on cross-type semantic alignment and collaborative multi-task anomaly optimization.

Method	Person			Event			Location			Event-Type		
	AUC	Macro-F1	Micro-F1	AUC	Macro-F1	Micro-F1	AUC	Macro-F1	Micro-F1	AUC	Macro-F1	Micro-F1
Social-Gover I												
HAN	0.7242	0.4946	0.7411	0.6321	0.4637	0.7945	0.7612	<u>0.5825</u>	0.7934	0.4513	0.4412	0.7895
HGT	0.7081	0.4767	0.6983	0.8207	0.5941	0.8607	0.7623	0.5679	0.8017	0.4857	0.4062	0.6842
HGRAN	0.8302	0.5874	0.7886	<u>0.8695</u>	0.5919	0.8576	0.7043	0.5616	<u>0.8174</u>	0.8333	0.6206	0.7847
HGATE	0.7154	0.4563	0.7055	0.7852	0.4971	0.7219	0.7236	0.4912	0.7603	0.6250	0.2017	0.2105
DOMINANT	<u>0.8397</u>	<u>0.7461</u>	<u>0.8314</u>	0.8005	0.6800	0.8562	0.8275	0.5475	0.8089	<u>0.8642</u>	0.7373	0.8521
CoLA	0.5604	0.4702	0.4893	0.5314	0.4681	<u>0.8646</u>	0.6710	0.4410	0.6281	0.5179	0.5064	0.5971
TFGAD	0.5361	0.4377	0.4727	0.6259	0.4514	0.5274	<u>0.9454</u>	0.4375	0.5537	0.5208	0.5128	0.5789
TAM	0.5653	0.4750	0.4917	0.5026	0.4659	0.8721	0.6739	0.4471	0.6116	0.5280	0.5128	0.5789
AEGIS	0.6706	0.6351	0.7150	0.7568	0.6131	0.8288	0.7830	0.4762	0.8091	0.7708	0.4571	<u>0.8351</u>
CSA-MTH	0.9094	0.8136	0.8480	0.9015	<u>0.6600</u>	0.6963	0.9565	0.6102	0.8347	0.9514	<u>0.6342</u>	0.7916
Social-Gover II												
HAN	0.6839	0.4727	0.7684	0.8701	0.4873	0.9323	0.5209	0.4516	0.8090	0.6235	0.4076	0.6938
HGT	0.6224	0.5409	<u>0.7984</u>	<u>0.9539</u>	<u>0.6928</u>	0.9694	0.5721	0.4663	0.8142	0.6291	0.4054	0.6818
HGRAN	<u>0.8625</u>	<u>0.5755</u>	0.7697	0.8940	0.5269	0.9375	0.7502	0.5642	<u>0.8483</u>	0.8055	<u>0.6319</u>	0.7831
HGATE	0.6372	0.4206	0.4143	0.6242	0.4848	0.7541	0.6488	0.4408	0.6556	0.6824	0.4058	0.5455
DOMINANT	0.7096	0.4441	0.7990	0.9346	0.4954	0.9817	0.8028	0.4617	0.8576	<u>0.9076</u>	0.4261	0.7424
CoLA	0.7110	0.5725	0.7711	0.7859	0.5594	0.9658	0.6844	0.4950	0.7962	0.7275	0.6182	<u>0.7869</u>
TFGAD	0.5843	0.4412	0.4912	0.9510	0.3728	0.5183	0.5313	0.4338	0.5064	0.5906	0.4849	0.5152
TAM	0.5165	0.3673	0.3700	0.7877	0.5937	<u>0.9724</u>	0.5082	0.3996	0.4510	0.7971	0.5968	0.6061
AEGIS	0.7020	0.5157	0.5359	0.4321	0.4914	0.8310	<u>0.9451</u>	0.7522	0.8244	0.9050	0.6169	0.6212
CSA-MTH	0.9225	0.6703	0.6724	0.9891	0.7517	0.9691	0.9567	<u>0.7245</u>	0.8201	0.9100	0.7657	0.7879

Table 2: AUC, Macro-F1 and Micro-F1 results with best bolded and the second best underlined.

4.3 Ablation Study

We design two ablation variants that correspond directly to the two core modules of CSA-MTHGAD:

- **w/o CSA.** Removes the entire semantic enhancement pipeline on text-less node types. Each node type is then encoded only from its raw attributes by the standard heterogeneous GNN. This variant tests whether injecting anomaly-sensitive textual cues from rich nodes is essential.
- **w/o MTH.** Disables the multi-task optimization mechanisms, training relies solely on uniform-weight reconstruction and cross-entropy losses. This variant evaluates the importance of mitigating gradient conflicts among heterogeneous node-type classification tasks.

Comparing CSA-MTHGAD with these two reduced versions quantifies (i) the gain from cross-type semantic alignment/feature enhancement and (ii) the gain from gradient-harmonized multi-task optimization.

We use AUC as the evaluation metric for the ablation study because this metric directly reflects the ranking quality of anomalous versus normal nodes. From Figure 3, **w/o CSA** is consistently and markedly worse than CSA-MTHGAD on every node type in both Social-Gover I and II (e.g., Location drops from 95.65% to 73.04% on Risk I; Event-Type drops from 95.14% to 91.67% on Social-Gover I).

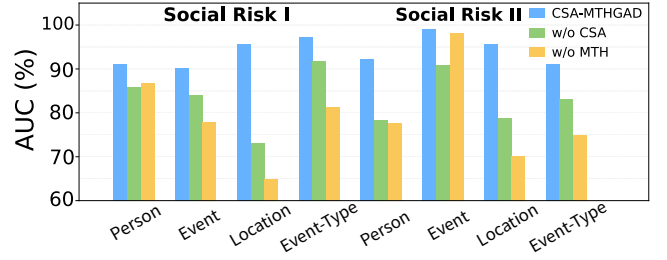


Figure 3: AUC of CSA-MTHGAD and its variants.

This verifies that the cross-type semantic alignment elevates the originally text-less node types (Person, Location, Event-Type) by supplying discriminative cues distilled from text-rich Event nodes. But its removal also causes a clear decline on the Event type itself, showing that simple heterogeneous propagation would otherwise introduce noisy backflow and dilute salient anomaly signals. Meanwhile, **w/o MTH** yields unbalanced degradation: some node types retain relatively higher AUC (e.g., Event on Social-Gover II) but others collapse sharply (e.g., Location from 95.65% to 64.78%, Event-Type from 97.14% to 81.25% on Social-Gover I), demonstrating that without mitigating gradient conflicts, optimization drifts toward a subset of tasks and sacrifices the rest. The full CSA-MTHGAD

achieves uniformly strong AUCs across all node types, confirming that (i) cross-type anomaly feature fusion and (ii) gradient-harmonized multi-task training are both indispensable and complementary for balanced, high-quality multi-type anomaly detection.

4.4 Hyper-Parameter Analysis

We investigate the impact of the hyper-parameter K on performance, which controls how many low-smoothness dimensions are selected from the text-rich nodes and subsequently injected into text-less nodes. K critically governs detection quality by mediating a fundamental tradeoff: sufficient information capture requires avoiding an undersized K that loses vital cues, while effective noise dilution demands preventing an oversized K that introduces redundant high-smoothness features. We therefore conduct a thorough study across $\text{keep_ratio} \in \{0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$. Figure 4 reports CSA-MTHGAD’s node-specific AUC results across Social-Gover I and Social-Gover II, covering four node types.

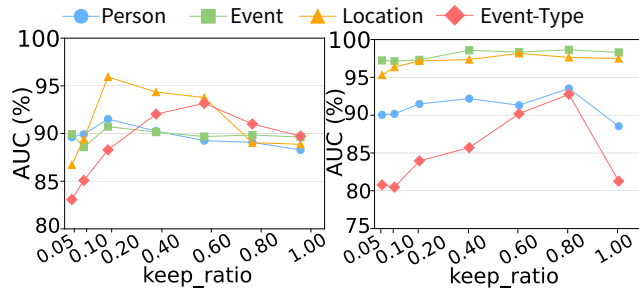


Figure 4: Effect of parameter K on the AUC

As shown in Figure 4, increasing the proportion of selected features produces a clear rise and fall trend in AUC for every node type on both Social-Gover I and Social-Gover II. When the ratio is too small, node representations, especially for the text-less nodes like Event-Type and Person categories lack sufficient information, leading to noticeably low AUC. With a moderate ratio, the model fully exploits the most discriminative, anomaly-sensitive dimensions while suppressing noise, yielding the highest and most balanced performance across all four node types. However, further enlarging the ratio introduces high-smoothness, redundant features that dilute anomaly cues: the gains of text-rich nodes saturate and the performance of text-less nodes declines. Hence, the intermediate feature ratio can achieve the best overall detection effect, which verifies the effectiveness and limited scope of the “less is more” low-smoothness feature selection strategy.

4.5 Visualization

To validate the effectiveness of the cross-type feature propagation mechanism and to improve the interpretability of the model’s decision-making process, we conducted a comparative t-SNE dimensionality reduction visualization on the Social-Gover II dataset. Figure 5 illustrates the spatial distribution of node representations both before and after the feature injection process, showcasing the differences in how the nodes are positioned in the embedding space. Before the semantic alignment via the CSA module, text-less nodes

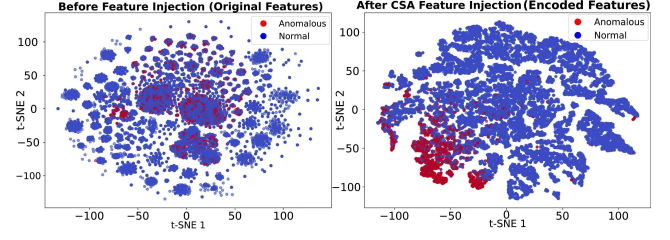


Figure 5: Visualization results

heavily relied on their own sparse feature sets and local structural information. As a result, the anomaly signals within these nodes were not adequately propagated, making it difficult to effectively separate anomalous nodes from normal ones. This resulted in a mixed distribution of anomalous and normal nodes within the embedding space, where the anomalies were scattered without clear separation boundaries.

After applying the CSA module, anomaly-sensitive features from text-rich nodes were selectively propagated to the text-less nodes. This semantic alignment significantly boosted the representation quality of the text-less nodes, allowing for a clearer distinction between normal and anomalous nodes. The updated feature led to a marked improvement in the cluster separation within the embedding space. Anomalous nodes now appeared as isolated regions, visually distinct from the dense clusters of normal nodes.

This t-SNE visualization directly confirms that the semantic alignment mechanism effectively mitigates the sparsity of anomaly signals in text-less nodes, thereby improving the detection capability for implicit abnormality and providing a reliable interpretable basis for the model’s decisions.

5 Conclusion

In this work, we presented CSA-MTHGAD, a unified framework for multi-type anomaly detection on heterogeneous social governance graphs. The model aligns cross-type semantics by selecting low-smoothness features from text-rich nodes and propagating them to text-less nodes. A two-layer heterogeneous GNN generates embeddings trained with attribute reconstruction, contrastive learning, and gated score propagation, while task-aware gradient coordination harmonizes objectives across node types. Extensive experiments on two real-world heterogeneous benchmarks show that CSA-MTHGAD consistently surpasses strong heterogeneous and homogeneous baselines in AUC and F1, with ablations and parameter analysis validating each design choice. We expect that the proposed framework and findings will open promising avenues for future research on more principled and interpretable multi-type anomaly detection in complex heterogeneous graphs.

Acknowledgments

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References

- [1] Aisha Abdallah, Mohd Aizaini Maarof, and Anazida Zainal. 2016. Fraud detection system: A survey. *J. Netw. Comput. Appl.* (2016), 90–113.
- [2] Leman Akoglu, Mary McGlohon, and Christos Faloutsos. 2010. oddball: Spotting Anomalies in Weighted Graphs. In *Advances in Knowledge Discovery and Data Mining, 14th Pacific-Asia Conference, PAKDD 2010, Hyderabad, India, June 21-24, 2010. Proceedings. Part II*. 410–421.
- [3] Sambaran Bandyopadhyay, Lokesh N, Saley Vishal Vivek, and M. Narasimha Murty. 2020. Outlier Resistant Unsupervised Deep Architectures for Attributed Network Embedding. In *WSDM '20: The Thirteenth ACM International Conference on Web Search and Data Mining, Houston, TX, USA, February 3-7, 2020*. 25–33.
- [4] Andrew P. Bradley. 1997. The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognit.* (1997), 1145–1159.
- [5] Bernardo Branco, Pedro Abreu, Ana Sofia Gomes, Mariana S. C. Almeida, João Tiago Ascensão, and Pedro Bizarro. 2020. Interleaved Sequence RNNs for Fraud Detection. In *KDD '20: The 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Virtual Event, CA, USA, August 23-27, 2020*. 3101–3109.
- [6] Kaize Ding, Jundong Li, Nitin Agarwal, and Huan Liu. 2020. Inductive Anomaly Detection on Attributed Networks. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI 2020*. 1288–1294.
- [7] Kaize Ding, Jundong Li, Rohit Bhanushali, and Huan Liu. 2019. Deep Anomaly Detection on Attributed Networks. In *Proceedings of the 2019 SIAM International Conference on Data Mining, SDM 2019, Calgary, Alberta, Canada, May 2-4, 2019*. 594–602.
- [8] Qi Ding, Natallia Katenka, Paul Barford, Eric D. Kolaczky, and Mark Crovella. 2012. Intrusion as (anti)social communication: characterization and detection. In *The 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '12, Beijing, China, August 12-16, 2012*. 886–894.
- [9] Haoyi Fan, Fengbin Zhang, and Zuoqiong Li. 2020. Anomalydae: Dual Autoencoder for Anomaly Detection on Attributed Networks. In *2020 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2020, Barcelona, Spain, May 4-8, 2020*. 5685–5689.
- [10] Xinyu Fu, Jiani Zhang, Ziqiao Meng, and Irwin King. 2020. MAGNN: Metapath Aggregated Graph Neural Network for Heterogeneous Graph Embedding. In *WWW '20: The Web Conference 2020, Taipei, Taiwan, April 20-24, 2020*. 2331–2341.
- [11] William L. Hamilton, Zitao Ying, and Jure Leskovec. 2017. Inductive Representation Learning on Large Graphs. In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*. 1024–1034.
- [12] Xueying Han, Susu Cui, Jian Qin, Song Liu, Bo Jiang, Cong Dong, Zhigang Lu, and Baoxu Liu. 2024. ContraMTD: An Unsupervised Malicious Network Traffic Detection Method based on Contrastive Learning. In *Proceedings of the ACM on Web Conference 2024, WWW 2024, Singapore, May 13-17, 2024*. 1680–1689.
- [13] Dongxiao He, Yiqi Dong, Xiaobao Wang, Meng Ge, Carl Yang, Di Jin, and Witold Pedrycz. 2025. Unveiling Fine-grained Deceptive Patterns in Multi-modal Fake News: An Explainable Neuro-Symbolic Framework with LVLMs. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025), 1–18.
- [14] Ziniu Hu, Yuxiao Dong, Kuansan Wang, and Yizhou Sun. 2020. Heterogeneous Graph Transformer. In *WWW '20: The Web Conference 2020, Taipei, Taiwan, April 20-24, 2020*. 2704–2710.
- [15] Di Jin, Jingyi Cao, Xiaobao Wang, Bingdao Feng, Dongxiao He, Longbiao Wang, and Jianwu Dang. 2025. Rethinking Contrastive Learning in Graph Anomaly Detection: A Clean-View Perspective. 3009–3017.
- [16] Di Jin, Jun Yang, Xiaobao Wang, Junwei Zhang, Shuqi Li, and Dongxiao He. 2025. A Dynamic Knowledge Update-Driven Model with Large Language Models for Fake News Detection. *ijcai.org*, 3000–3008.
- [17] Yawen Li, Xiaobao Wang, Bin Wen, Di Jin, and Junping Du. 2026. Sentiment Variation-Aware Sentiment Spike Explanation During COVID-19 Epidemic. *IEEE Transactions on Knowledge and Data Engineering* (2026), 1306–1318.
- [18] Ninghao Liu, Xiao Huang, and Xia Hu. 2017. Accelerated Local Anomaly Detection via Resolving Attributed Networks. In *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence, IJCAI 2017, Melbourne, Australia, August 19-25, 2017*. 2337–2343.
- [19] Yixin Liu, Shiyuan Li, Yu Zheng, Qingfeng Chen, Chengqi Zhang, and Shirui Pan. 2024. ARC: A Generalist Graph Anomaly Detector with In-Context Learning. In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- [20] Yixin Liu, Zhao Li, Shirui Pan, Chen Gong, Chuan Zhou, and George Karypis. 2022. Anomaly Detection on Attributed Networks via Contrastive Self-Supervised Learning. *IEEE Trans. Neural Networks Learn. Syst.* (2022), 2378–2392.
- [21] Xuexiong Luo, Jia Wu, Amin Beheshti, Jian Yang, Xiankun Zhang, Yuan Wang, and Shan Xue. 2022. ComGA: Community-Aware Attributed Graph Anomaly Detection. In *WSDM '22: The Fifteenth ACM International Conference on Web Search and Data Mining, Virtual Event / Tempe, AZ, USA, February 21 - 25, 2022*. 657–665.
- [22] Hezhe Qiao and Guansong Pang. 2023. Truncated Affinity Maximization: One-class Homophily Modeling for Graph Anomaly Detection. In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- [23] Pin Ren, Yan Gao, Zhichun Li, Yan Chen, and Benjamin Watson. 2006. IDGraphs: Intrusion Detection and Analysis Using Stream Compositing. *IEEE Computer Graphics and Applications* (2006), 28–39.
- [24] Heng Tao Shen. 2009. Principal Component Analysis. In *Encyclopedia of Database Systems*. Springer US, 2136.
- [25] Andrés Troya-Galvis, Pierre Gançarski, and Laure Berti-Équille. 2018. Remote sensing image analysis by aggregation of segmentation-classification collaborative agents. *Pattern Recognit.* (2018), 259–274.
- [26] Petar Velickovic, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph Attention Networks. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net.
- [27] Minjie Wang, Lingfan Yu, Da Zheng, Quan Gan, Yu Gai, Zihao Ye, Mufei Li, Jinjing Zhou, Qi Huang, Chao Ma, Ziyue Huang, Qipeng Guo, Hao Zhang, Haibin Lin, Junbo Zhao, Jinyang Li, Alexander J. Smola, and Zheng Zhang. 2019. Deep Graph Library: Towards Efficient and Scalable Deep Learning on Graphs. *CoRR* (2019).
- [28] Wei Wang, Xiaoyang Suo, Xiangyu Wei, Bin Wang, Hao Wang, Hong-Ning Dai, and Xiangliang Zhang. 2023. HGATE: Heterogeneous Graph Attention Auto-Encoders. *IEEE Trans. Knowl. Data Eng.* (2023), 3938–3951.
- [29] Xiang Wang, Weikang Deng, Zhenyu Meng, and Dewang Chen. 2024. Hybrid-attention mechanism based heterogeneous graph representation learning. *Expert Syst. Appl.* (2024), 123963.
- [30] Xiaobao Wang, Yiqi Dong, Di Jin, Yawen Li, Longbiao Wang, and Jianwu Dang. 2023. Augmenting Affective Dependency Graph via Iterative Incongruity Graph Learning for Sarcasm Detection. 4702–4710.
- [31] Xiaobao Wang, Meng Ge, Lingshan Li, Di Jin, Kai He, and Erik Cambria. 2025. Global Entity Relationship Enhancement Network for Multimodal Sarcasm Detection. *IEEE Transactions on Affective Computing* (2025), 1–14.
- [32] Xiao Wang, Houye Ji, Chuan Shi, Bai Wang, Yanfang Ye, Peng Cui, and Philip S. Yu. 2019. Heterogeneous Graph Attention Network. In *The World Wide Web Conference, WWW 2019, San Francisco, CA, USA, May 13-17, 2019*. 2022–2032.
- [33] Xuhong Wang, Baihong Jin, Ying Du, Ping Cui, Yingshui Tan, and Yupu Yang. 2021. One-class graph neural networks for anomaly detection in attributed networks. *Neural Comput. Appl.* (2021), 12073–12085.
- [34] Xiaobao Wang, Ruoxiao Sun, Yujun Zhang, Bingdao Feng, Dongxiao He, Luzhi Wang, and Di Jin. 2025. Stealthy Yet Effective: Distribution-Preserving Backdoor Attacks on Graph Classification. *CoRR abs/2509.26032* (2025).
- [35] Xiaobao Wang, Yujing Wang, Dongxiao He, Zhe Yu, Yawen Li, Longbiao Wang, Jianwu Dang, and Di Jin. 2025. Elevating Knowledge-Enhanced Entity and Relationship Understanding for Sarcasm Detection. *IEEE Trans. Knowl. Data Eng.* (2025), 3356–3371.
- [36] Yuchen Wang, Jinghui Zhang, Zhengjie Huang, Weibin Li, Shikun Feng, Ziheng Ma, Yu Sun, Dianhai Yu, Fang Dong, Jiahui Jin, Beilun Wang, and Junzhou Luo. 2023. Label Information Enhanced Fraud Detection against Low Homophily in Graphs. In *Proceedings of the ACM Web Conference 2023, WWW 2023, Austin, TX, USA, 30 April 2023 - 4 May 2023*. 406–416.
- [37] Shujie Yang, Binchi Zhang, Shangbin Feng, Zhaoxuan Tan, Qinghua Zheng, Jun Zhou, and Minnan Luo. 2022. AHEAD: A Triple Attention Based Heterogeneous Graph Anomaly Detection Approach. *CoRR* (2022).
- [38] Zhizhi Yu, Chungong Liang, Xinglong Chang, Dongxiao He, Di Jin, and Jianguo Wei. 2025. Dynamic Neighborhood Modeling via Node-Subgraph Contrastive Learning for Graph-Based Fraud Detection. In *AAAI-25, Sponsored by the Association for the Advancement of Artificial Intelligence, February 25 - March 4, 2025, Philadelphia, PA, USA*. 13115–13123.
- [39] Jintao Zhang and Quan Xu. 2021. Attention-aware heterogeneous graph neural network. *Big Data Min. Anal.* (2021), 233–241.
- [40] Yu Zheng, Ming Jin, Yixin Liu, Lianhua Chi, Khoa Tran Phan, and Yi-Ping Phoebe Chen. 2023. Generative and Contrastive Self-Supervised Learning for Graph Anomaly Detection. *IEEE Trans. Knowl. Data Eng.* (2023), 12220–12233.
- [41] Cheng Zhou, Guangxia Li, Hao Weng, and Yiyu Xiang. 2025. Training-free Graph Anomaly Detection: A Simple Approach via Singular Value Decomposition. In *Proceedings of the ACM on Web Conference 2025, WWW 2025, Sydney, NSW, Australia, 28 April 2025 - 2 May 2025*. 4196–4205.
- [42] Yanqiao Zhu, Yichen Xu, Hejie Cui, Carl Yang, Qiang Liu, and Shu Wu. 2022. Structure-Enhanced Heterogeneous Graph Contrastive Learning. In *Proceedings of the 2022 SIAM International Conference on Data Mining, SDM 2022, Alexandria, VA, USA, April 28-30, 2022*. 82–90.

A Related Work

A.1 Graph Anomaly Detection

Graph structured data has gained increasing attention in anomaly detection due to its ability to capture complex relational dependencies. Significant advancements have been made in detecting anomalies on graph structured data. Traditional non-deep methods often reformulate graph anomaly detection as a conventional anomaly detection task. This is necessary because conventional methods are designed solely for tabular data and cannot inherently process graph-structured data with its complex structural information. To overcome this limitation, numerous approaches[2, 8, 18] have utilized node level statistical features, such as in-degree and out-degree, to identify anomalous nodes. The rise of Graph Neural Networks (GNNs) has spurred the development of numerous novel techniques for detecting anomalies on graphs. DOMINANT[7] employs a GCN-based encoder-decoder architecture to jointly reconstruct node attributes and graph structure for detecting anomalies. CoLA[20] introduces contrastive learning to model the normality between nodes and their local subgraphs, thereby enabling effective separation of anomalous nodes. AEGIS[6] proposes an inductive framework that integrates graph differencing with adversarial learning, enabling efficient detection of newly arriving abnormal nodes in graphs. TAM[22] leverages a one-class homogeneity assumption and maximizes local node affinity over truncated graph structures to enhance unsupervised anomaly detection. While these methods have achieved promising results on homogeneous graphs, anomaly detection on heterogeneous graphs remains underexplored. Existing approaches often fail to fully leverage the rich semantic and structural heterogeneity, underscoring the need for more tailored and generalizable solutions in heterogeneous settings.

A.2 Heterogeneous graph neural network

Heterogeneous graph neural network has garnered significant attention due to its crucial role in modeling complex relationships. One class of methods employs meta-paths to model complex relationships within heterogeneous graph data. HAN[32] proposed a meta-path-based structural and semantic attention mechanism aimed at learning node representations that capture higher-order semantic relationships within heterogeneous information networks. MAGNN[10] further integrates various semantic information at the meta-path level by performing intermediate node-aware attention aggregation on node sequences within meta-paths in heterogeneous graphs. Another class of methods abandons meta-paths. HGT[14] designs a node-type and edge-type-aware attention mechanism to model complex semantic dependencies in heterogeneous graphs within the Transformer framework. AHNN[39] proposes a multi-level attention mechanism that automatically extracts semantic information from heterogeneous graph structures through node-type and relationship-type-aware aggregation. While these methods have shown considerable success in modeling relationships in heterogeneous graphs, many challenges remain in fully capturing the complex and diverse interactions between different node and edge types. Existing approaches often fail to effectively exploit the full potential of semantic and structural heterogeneity, highlighting the need for more adaptive and robust solutions that can scale across varied heterogeneous graph scenarios.

B Exploring Multi-Task Optimization Strategies

To further examine the role of task-weighting in the multi-type classification objective of CSA-MTHGAD, we investigate several alternative weighting schemes for the multi-task loss. CSA-MTHGAD performs anomaly detection over four node types (Person, Event, Location, and Event-Type) under a unified training objective, where the multi-task classification term aggregates type-specific cross-entropy losses as $\tilde{\mathcal{L}}_{\text{cls}} = \sum_{t \in \mathcal{T}} \pi_t \mathcal{L}_{\text{cls}}^t$. In the original implementation, π_t is dynamically determined by jointly considering task difficulty (measured by the mean loss) and task scale (measured by the number of training nodes), in order to prevent the optimization from being dominated by any single node type.

We compare three task-weighting variants while keeping the model architecture, training protocol, and hyper-parameters unchanged:

- **Loss-based weighting (Change1).** $\pi_t \propto \tilde{\mathcal{L}}_{\text{cls}}^t$, where $\tilde{\mathcal{L}}_{\text{cls}}^t$ denotes the mean classification loss of task t in the current epoch. This strategy prioritizes tasks that are currently harder to optimize.
- **Size-based weighting (Change2).** $\pi_t \propto 1/\sqrt{n_t}$, where n_t is the number of training nodes of type t . This re-balances supervision by up-weighting under-represented node types.
- **Mixed weighting (Change3).** With a mixing factor α (set to 0.5), $\pi_t \propto (\tilde{\mathcal{L}}_{\text{cls}}^t)^\alpha (1/\sqrt{n_t})^{1-\alpha}$, followed by normalization such that $\sum_{t \in \mathcal{T}} \pi_t \approx |\mathcal{T}|$.

Figure 6 reports the AUC of the three variants and the original model on Social-Gover II, evaluated separately on the four node types. Overall, the original weighting strategy achieves the most stable and consistently strong performance across all types, suggesting that jointly accounting for both task difficulty and task scale provides a more reliable coordination signal than using either factor alone.

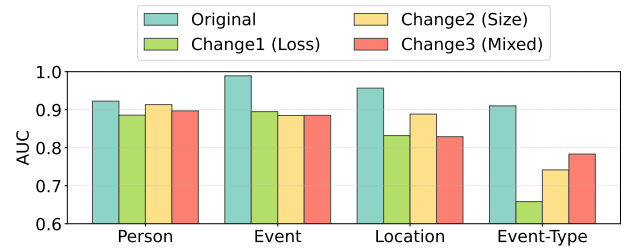


Figure 6: AUC of different multi-task weighting strategies on four node types.

Two observations can be drawn from the results. First, **single-factor weighting may lead to unstable optimization across tasks**. Loss-based weighting (Change1) causes a noticeable drop in Event-Type AUC and also degrades Location performance, indicating that aggressively emphasizing high-loss tasks can bias learning toward noisier or more difficult supervision signals, thereby weakening generalization on other types. Size-based weighting (Change2) yields improvements over Change1 on Person and Location, but still underperforms the original strategy on Event and

Event-Type, implying that correcting data imbalance alone does not fully resolve cross-task conflicts in joint optimization. Second, **a naive mixture does not necessarily recover the best coordination effect**. Although Change3 improves Event-Type AUC compared with Change1/2, it remains inferior to the original method on Event and Location, highlighting that effective multi-task coordination requires careful calibration of task weights rather than a direct interpolation of difficulty- and size-based heuristics. In summary, the original task-weighting strategy provides the most robust performance across node types in our setting.

C LIMITATIONS AND DISCUSSION

Although CSA-MTHGAD demonstrates strong performance in multi-type heterogeneous graph anomaly detection, there remain several directions that warrant further exploration. First, the experiments of CSA-MTHGAD were conducted only on two heterogeneous graph datasets in the domain of social governance. While this setting sufficiently reflects the applicability of the model in real-world scenarios, it may also be constrained by the lack of comparisons with other benchmarks. This limitation primarily arises from the sensitivity and confidentiality of data in the field of social governance. At present, no publicly available heterogeneous graph datasets provide annotations aligned with our problem setting for multi-type anomaly detection in a secure manner, and

thus we were unable to extend the experiments to other public benchmarks. We hope this work can inspire future researchers and practitioners to construct more benchmarks tailored to heterogeneous graph anomaly detection in social governance. In addition, with respect to multi-task learning and anomaly signal propagation, CSA-MTHGAD is capable of effectively transmitting cross-type anomaly information and achieves superior performance in multi-type anomaly detection within social governance. However, in more complex scenarios anomaly signals may appear more implicit or exhibit stronger nonlinearity, and the current method may not fully capture such complex patterns. Furthermore, the mechanisms for cross-type feature selection and information fusion may still be limited when tasks are highly imbalanced or when performance depends on long-range dependencies. Future research may further explore more refined feature learning mechanisms and adaptive cross-type signal fusion strategies to enhance the model's effectiveness and generalization in highly complex tasks and cross-domain applications.

Despite these research directions, CSA-MTHGAD represents the first attempt to provide an effective and interpretable multi-task framework for heterogeneous graph anomaly detection in the domain of social governance. We believe this work not only offers a novel method for social governance tasks but also provides new perspectives and potential for subsequent research.